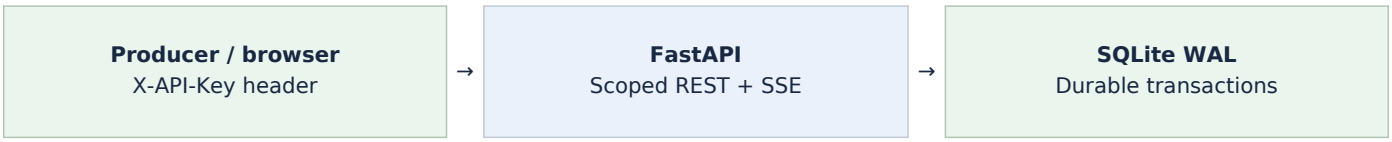
Design before code — event integrity and isolation

One-page architecture | Source and expanded discussion: DESIGN.md



01 / Event format and ordering

POST /runs/{id}/metrics: {batch_id: UUID, events: [{event_id: UUID, name, value: finite float, observed_at: ISO-8601 timezone}]}. Each commit also stores received_at and a per-run integer seq. Writers use SQLite BEGIN IMMEDIATE and update next_seq in the same transaction as inserted metric rows. seq is commit order, never client timestamp order. A late observed_at is not reordered into earlier commits.

02 / Identical and conflicting retries

The pair (run_id, batch_id) stores a canonical request digest plus original response. A bit-for-bit equivalent retry returns its original acknowledgement with zero new rows. A changed batch with the same ID returns 409. In different batches an unchanged event_id is deduped; a changed payload with the same ID returns 409, rolling back the whole transaction. Keys must remain stable across network retries.

03 / Unreliable network and catch-up

On initial load, request paginated metrics after seq=0. On stream connect and reconnect, send Last-Event-ID with the last **applied** seq; the server emits id: seq and event: metric in order. The UI deduplicates by seq. Durable DB paging of 100 at a time replaces in-memory subscriber queues, so a slow reader gets backpressure rather than silent queue eviction. There is no configured retention pruning.

04 / Project isolation

Each project receives a 256-bit random API key; only its SHA-256 hash is stored. Every authenticated request resolves the project from the key. Every run, history, write and stream query constrains both project_id and run_id. Unknown and wrong-tenant IDs return identical 404 responses; UUIDv4 run IDs are impractical to enumerate. Key rotation revokes the presented key atomically. No response cache; authenticated responses specify no-store. Keys never enter URLs.

05 / Silent failures to probe

Disconnect/replay loss: compare SSE sequences with DB after restoring connectivity. Duplicate deliveries: retry the identical batch and same event in a new batch. Slow readers: paginate 570 records in groups of 17, and test 50 live subscribers. Partial writes: insert then cause a conflicting event, verify rollback. Rotation: assert the old key yields 401 on read/write/SSE. Logs, timing, proxy buffering and power loss are residual risks.

06 / Production decision

DON'T SHIP for 50 × 10 events/s as a production guarantee. A 4-round local 500 events/s pulse passed with 50 streams and 2,000 DB-matched deliveries. Still needed: unrestricted browser E2E, Docker smoke, rate limits, quota and audit controls, multi-instance/PostgreSQL design, delivery-lag telemetry and long-running proxy-backed soak tests.